

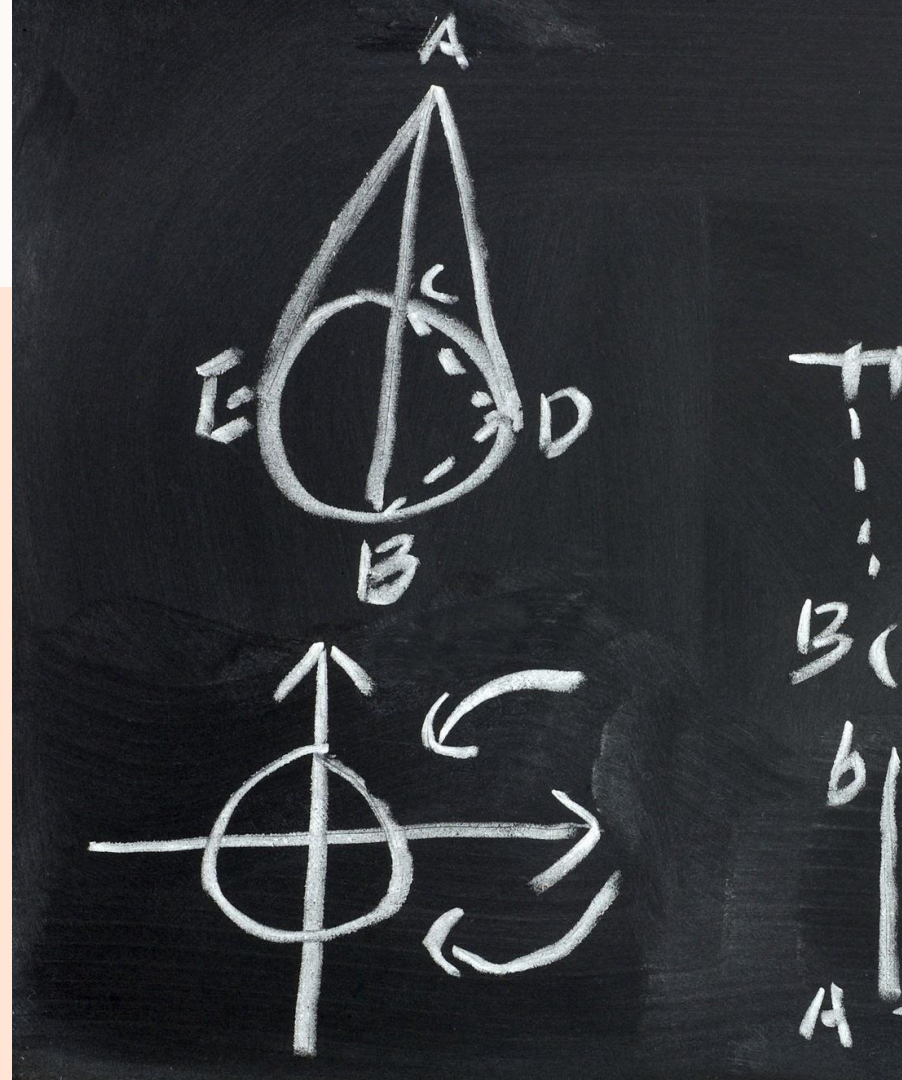


Steam Turbine Project

Objectives

For this experiment, our group aimed to:

- Create a steam turbine.
- Observe the efficiency between the input energy from a burner (thermal) and the output energy from the spinning fan (rotational kinetic).



Setup

Our material list:

- Hot plate set to 550 C
- Aluminum can with small hole
- Constructed turbine apparatus with reflective tape
- Tachometer
- ~20mL of water



Media

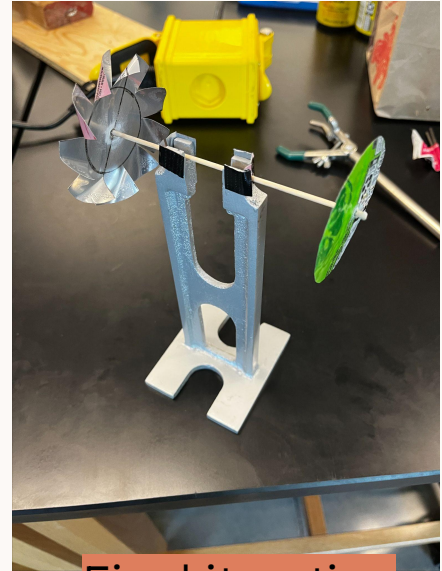
[Link to the Final Iteration Execution](#)



Earlier iteration



Earlier iteration



Final iteration

Results and Challenges

Results

- Data collected was low quality and it was unable to be processed effectively. (couldn't satisfy $\eta = \frac{P_{\text{out}}}{P_{\text{in}}}$)
- We were able to get the fan spinning and collect a rough estimate of how fast it spun.

Challenges and Iterations

- To calculate the amount of energy delivered to the boiler from the hot plate, we would need the area of the boiler contacting the hot plate, and the contact thermal resistance (each very difficult to measure).
- Even if all of this was known, the calculated efficiency would likely be extraordinarily low due to: friction losses, thermal energy loss to the air, imperfect steam contact angles, and many other factors.

Results and Challenges

Results

- Data collected was low quality and it was unable to be processed effectively. (couldn't satisfy $\eta = \frac{P_{\text{out}}}{P_{\text{in}}}$)
- We were able to get the fan spinning and collect a rough estimate of how fast it spun.

Challenges and Iterations

- We discussed using an ethanol burner, but it was not easily sourceable (a rubbing alcohol burner and candles both would not work).
- Construction of the apparatus had to be extremely precise in order to work without human interaction.
- Fan blades had to be reworked.
- Tachometer didn't work well with our setup.

Possible Errors

Errors

- Wind turbine placed at an angle.
- The temperature not being at exactly 550 as shown on the heating plate.
- Tachometer not accurately showing the rpm (reflective tape vs reflective aluminum).
- Condensation build-up on the fan blades.
- Hole not being perfectly centered.
- Steam shooting out at an unwanted angle.

How we minimized them

- Replaying the video of the wind turbine and manually counting and averaging the rpm.
- Do the experiment twice, recording the water on the second trial.

Possible Future Work

- 1) Power an LED using the rotation of the fan blades. (see Figure 1.1)
- 2) Create a more efficient system by minimizing energy loss (whether that be through friction or heat loss).
- 3) If we go with (2) our calculations may be much easier to compute.
- 4) If we had the funds, we could upgrade our materials to help us make our calculations more accurate.

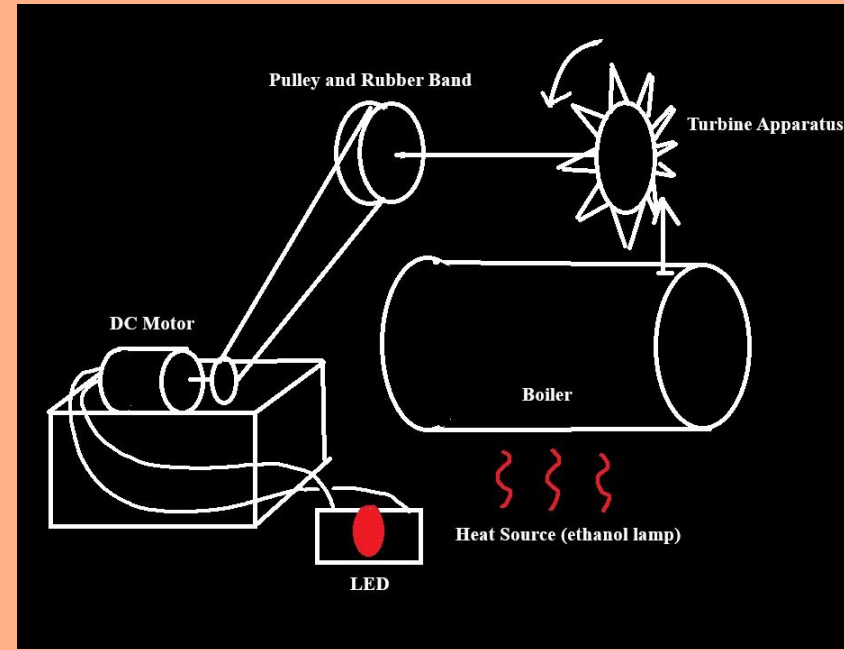


Figure 1.1 - Ideal self supported experimental setup

Summary

We were able to:

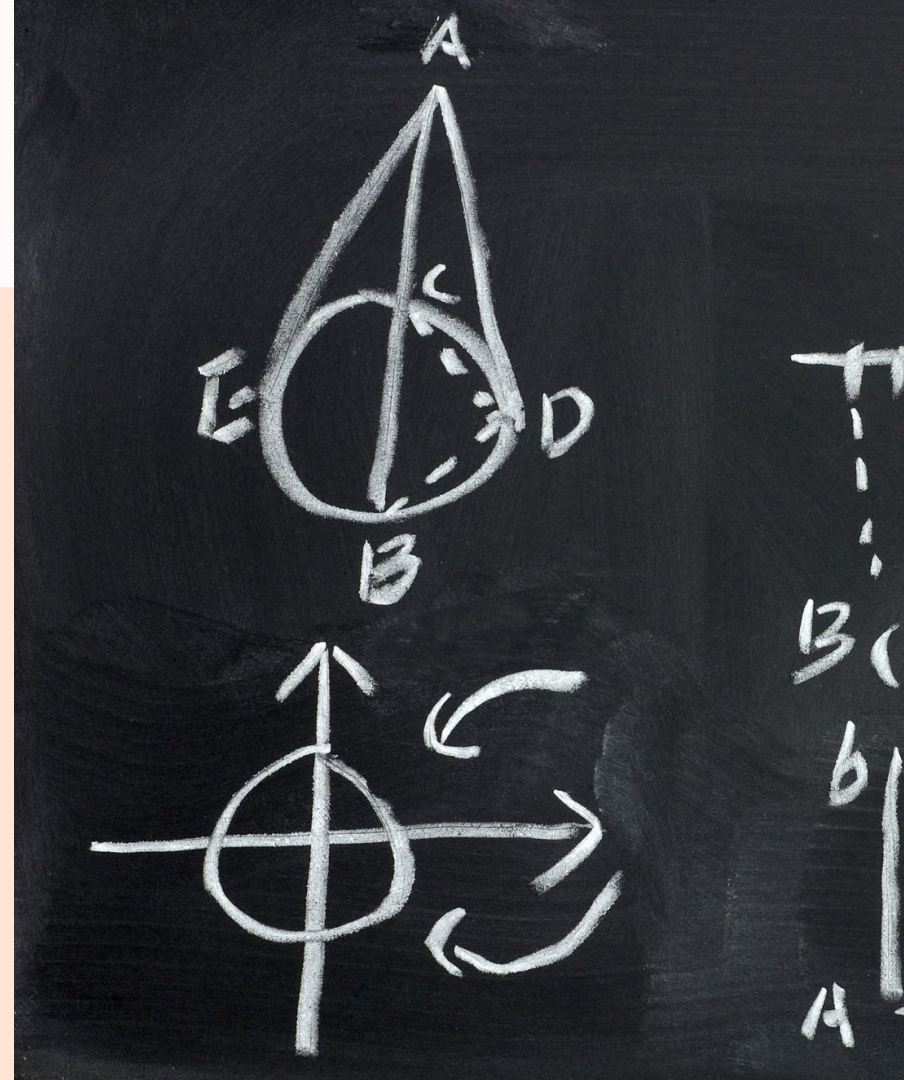
- Create a functional heat turbine engine.
- Calculate a rough estimate of how fast the fan spun.

Some challenges were:

- Difficult computations.
- Several iterations.

In the future:

- Turn on an LED.
- Maximize efficiency.
- Better our calculations.





Thank you!